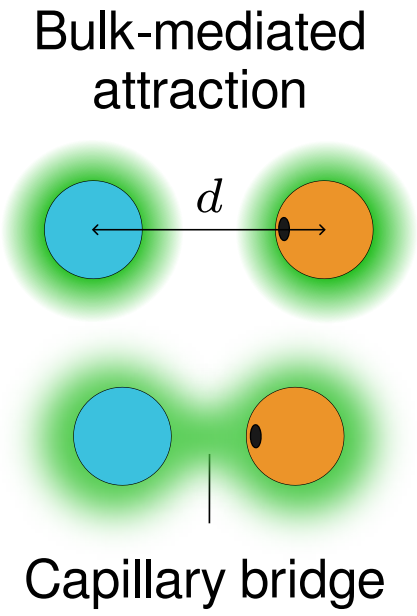

Cahn-Hilliard total energy

$$F = \int_B \left[\frac{\kappa}{2}(\nabla\phi)^2 + f(\phi)\right] d^3x - \int_{S_1} h_0\phi(\mathbf{x})dS - \int_{S_2} h_0\phi(\mathbf{x})dS$$